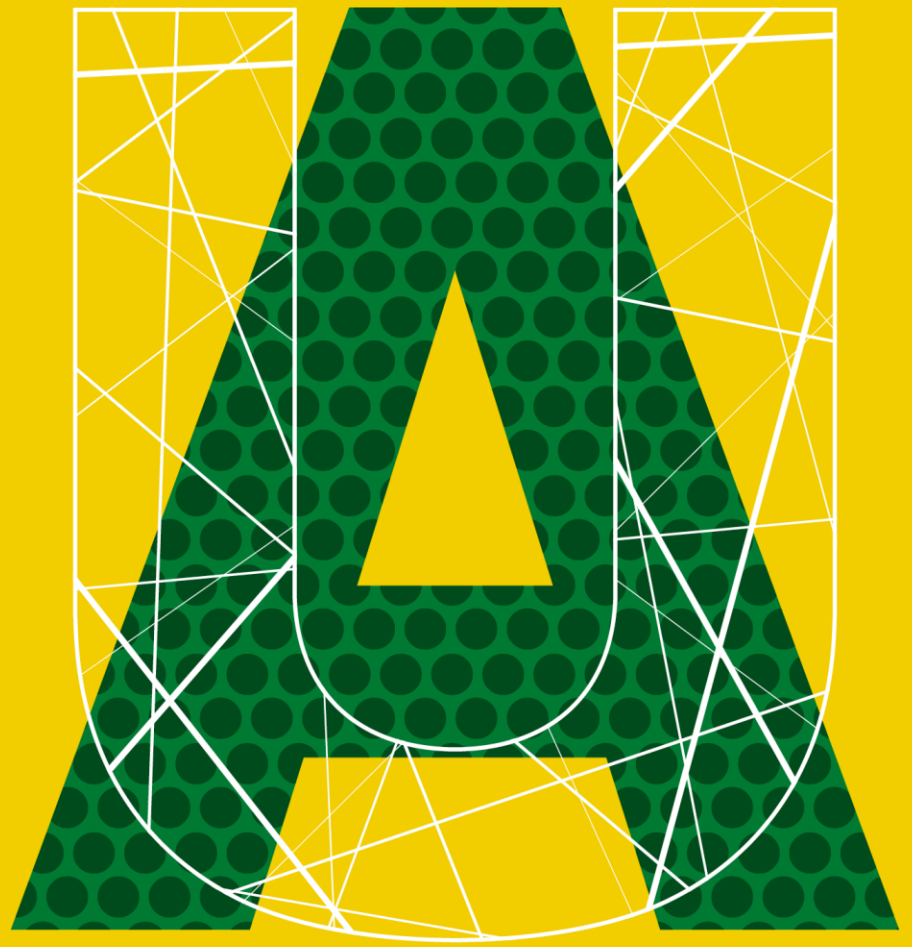
Efficient Robotic Pushing with Diffusion Policy:

Adaptive Execution and Self-Imitation-Based Fine-Tuning

Qihang Feng

UNIVERSITY OF ALBERTA

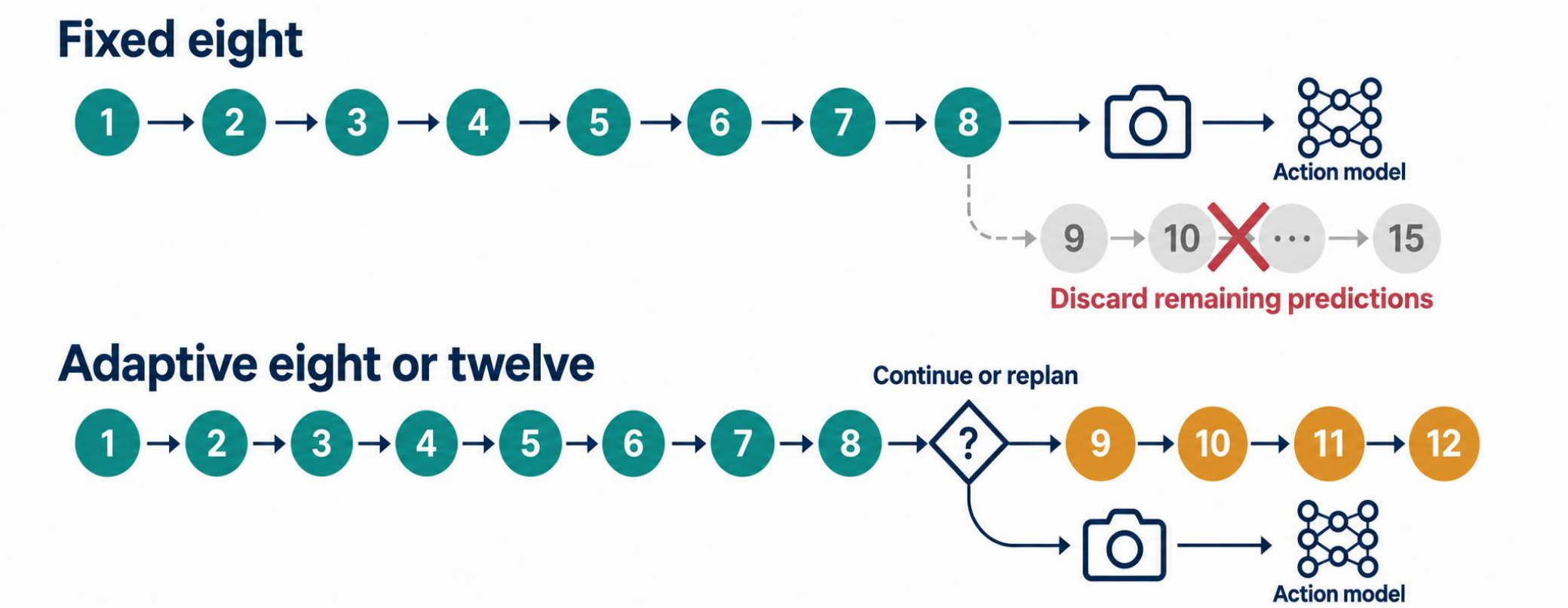


ECE 730 PROJECT PROPOSAL

Computation and AI for Cyber-Physical Systems

Prediction and execution

Each prediction requires computation. Remaining movements may still be useful.



The model generates the same sequence length in both cases

Signals for continuing or replanning

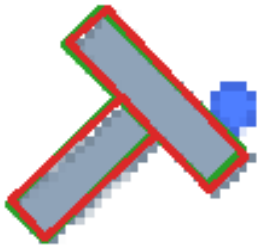
Signals: target coverage, decline, recovery and path smoothness

Continue up to twelve

limited recent decline, and smooth
remaining movements

Replan after eight

Near target, or a failed continuation check



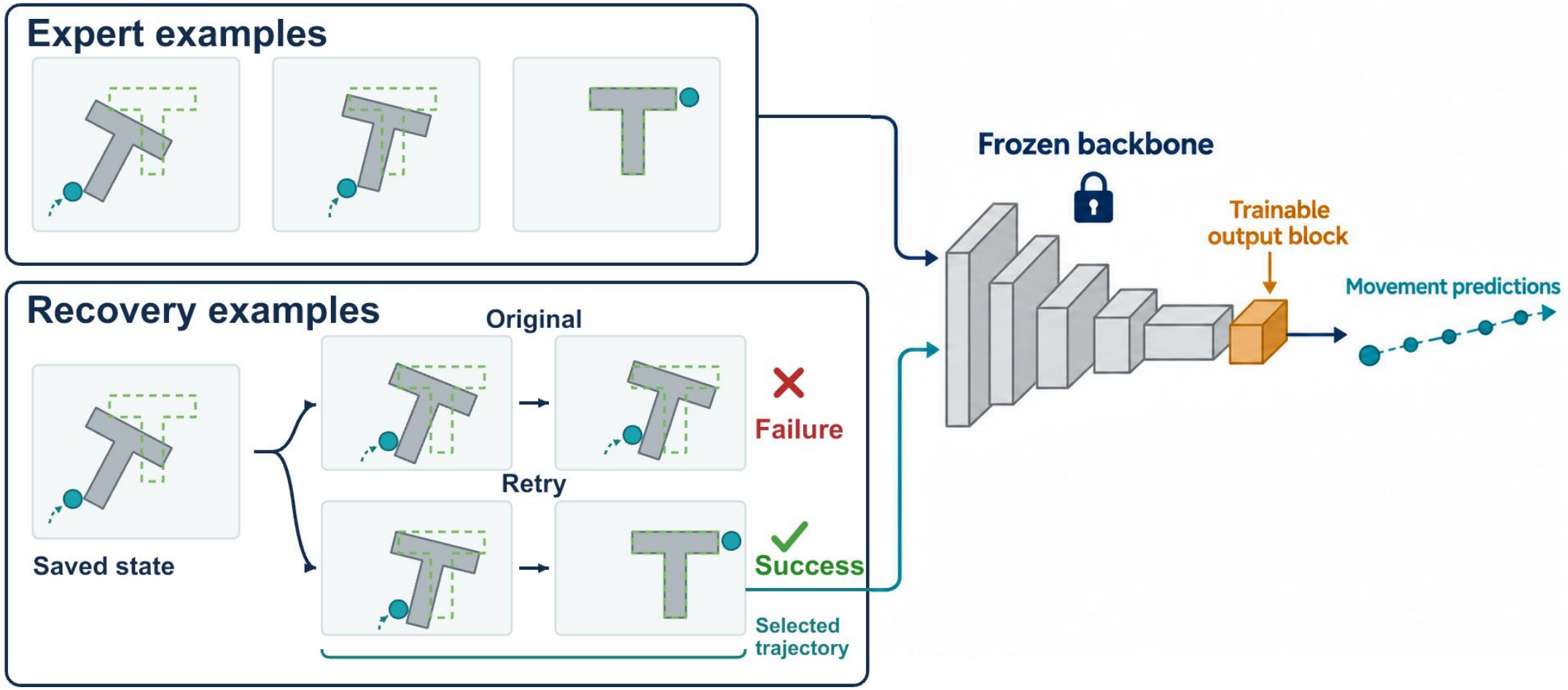
Interrupt early

Repeated warnings, accumulated alignment loss,
and no clear recovery

A temporary alignment drop can be part of a useful adjustment

Self-imitation-based fine-tuning

Retry saved states from failed cases and retain successful continuations.



Update only the final convolution block. Keep other weights frozen.

Planned evaluation

Configuration	Model	Execution
Original baseline	Original pretrained	Fixed eight
Adaptive execution	Original pretrained	Adaptive 8/12
Fine-tuning	Fine-tuned	Fixed eight
Adaptive execution + fine-tuning	Fine-tuned	Adaptive 8/12

Matched scenes, repeated prediction samples, equal movement limits

Success (>95% coverage) and average final alignment
Average prediction calls and average complete-attempt runtime

Expected contribution and scope

Target outcome

More successful pushes with fewer predictions and shorter runtime

Assess the effect on average final alignment

Report improvements, regressions and uncertainty

Scope: 2D PushT simulation on my laptop; no physical-robot or real-time deployment
Simulation pauses while the model predicts movements

Thank you!



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